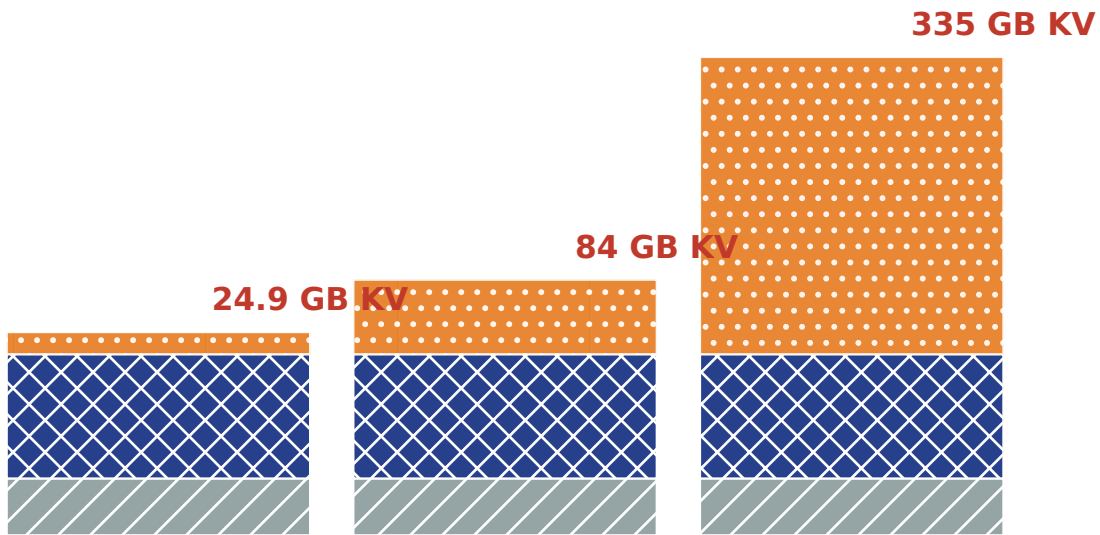


Memory Tetris: how the 8×H100 host (640 GB) fills with context

640 GB pool = 8×H100 HBM



9.5K max

= 229 GB used

32K context

= 288 GB used

128K context

= 539 GB used

baseline (weights + runtime) is context-independent; the KV cache is the lever that grows with context. FP16 ~2.5 MB/token [ILLUSTRATIVE][DERIVED].

 KV cache (grows w/ context)

 weights 140 GB

 runtime / NCCL ~64 GB